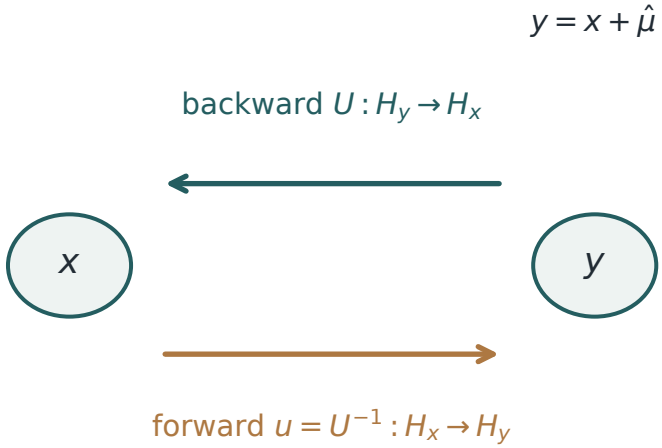
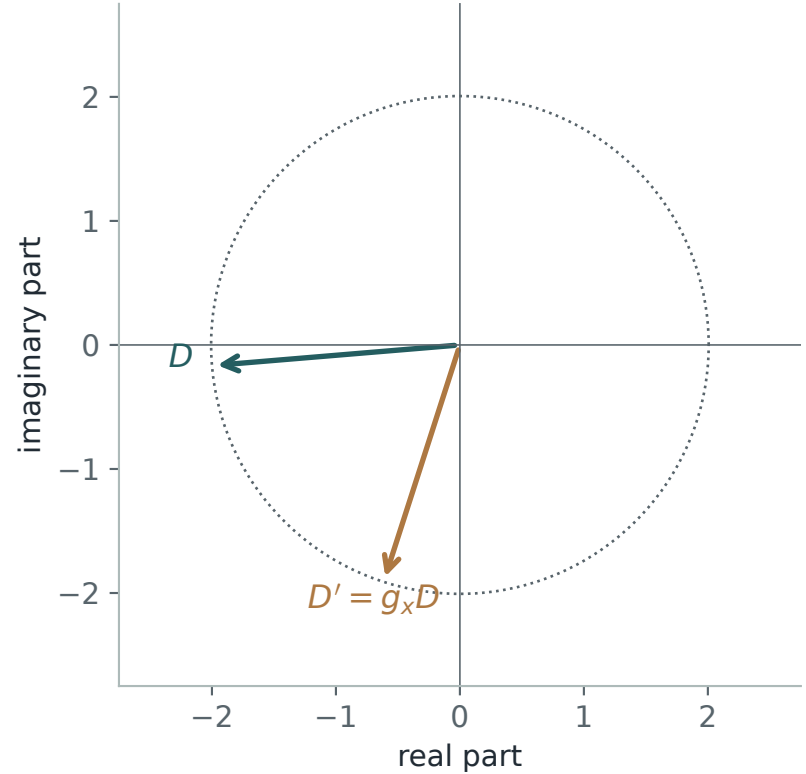
Transport direction fixes the endpoint factors



$U' = g_x U g_y^{-1} \qquad u' = g_y u g_x^{-1}$

A gauge change rotates the covariant difference